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Magnetization of the Austell Gneiss, Northwest Georgia Piedmont

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The Austell gneiss is located in the Piedmont 16 km west of Atlanta, Georgia, and 8 km northwest of the Brevard fault zone. This metamorphosed granite gneiss lies refolded in the nose of the Austell-Frolona antiform, a doubly plunging structure of regional extent. Subsequent to intrusion, the unit has undergone four episodes of deformation. The first and second generation folds (F_1 and F_2) are coaxial but not coplanar. The two later fold events were less intense and have little effect on local outcrop patterns in the study area. The age of the unit has been estimated to be greater than 325 m.y. but less than 365 m.y. The induced and remanent magnetic properties for at least six cores drilled at each of 10 sites within the Austell gneiss have been determined. The magnetic foliation for the unit is approximately parallel with field estimates for S_1 foliation planes exhibited by mineral orientations within the body, indicating post-emplacement magnetic grain mobility. The stable remanent magnetization exhibited by the body was therefore probably acquired during or slightly after the F_1 or F_2 deformational events as a result of cooling from metamorphic temperatures (~600°C), or, alternatively, from secondary magnetite grain nucleation and growth. A paleopole position calculated for these data without structural correction, ϕ=126.2, θ=34.4, dp=6.6, dm=13.2, can be explained by magnetic acquisition in at least three possible geologic settings. From simple to complex, these are the following: First, since a large oval of confidence is observed for the unit and minor tilting can not be resolved, the data may be explained by magnetization of the Austell at its present site during cooling from metamorphic temperatures. Second, a northwestward thrust event may have resulted in bringing the Austell to its present location. Third, the data can be explained by two major translational events. The Austell may have initially been magnetized and then moved to the northeast by transcurrent strike slip faulting >1000 km into juxtaposition with the southern Appalachians, followed by thrusting of the unit to the northwest to its present location. In support of this last interpretation is the similarity between the paleopole for the Austell and paleopoles determined for the Maritime Provinces of North America, which were suggested to have moved ~1500 km to the northeast during the mid to late Paleozoic. Against such an interpretation is the lack of geologic evidence in support of large-scale left-lateral strike slip faulting in the southern Appalachians.

INTRODUCTION

There are a large number of Paleozoic igneous and metamorphic units exposed in the southern Appalachian Piedmont. Only a few of these, such as the Elberton granite in Georgia, have been extensively studied [e.g., *Ellwood and Whitney*, 1980; *Ellwood et al.*, 1980; *Wenner*, 1981; *Stormer and Whitney*, 1981] primarily because they are poorly exposed due to high rates of weathering and low relief in the southern Piedmont. Further, the controls on the magnetization of metamorphic rocks are not well understood. Although an evaluation of the effects of weathering on the magnetization of a granitic body has been reported [*Ellwood*, 1981] as well as a number of magnetic results for relatively fresh granitic rocks in the region [*Ellwood and Wenner*, 1981; *Brown and Barton*, 1980], the magnetization of the gneissic units in the southeast is relatively unknown. Many of these gneissic units are present in the southern Piedmont. Some may potentially yield useful paleomagnetic information because of the presence of a number of excellent exposures in road cuts and quarries and because the geology is relatively well known [e.g., *Abrams and McConnell*, 1981*a*].

GEOLOGIC SETTING

The Austell gneiss has been variously described as a granite [*Crickmay*, 1952; C. W. Hayes, unpublished map] a granite augen gneiss [*Schepis*, 1952], and a gneiss [*Medlin and Crawford*, 1973]. Following several studies [*Higgins*, 1966, 1968; *Crawford and Medlin*, 1973, 1974; *McConnell and Abrams*, 1978], *Abrams and McConnell* [1981*a*] have made the Austell gneiss a formal stratigraphic unit.

The Austell gneiss is located within the northern Piedmont of Georgia, 16 km west of Atlanta and 8 km northwest of the Brevard fault zone (Figure 1). The body, an augen textured orthogneiss, is granitic in composition. Of the accessory minerals observed in the unit, only magnetite, ilmenohematite, and hematite are magnetically significant.

Metamorphism and folding. The Austell gneiss lies within the doubly plunging Austell-Frolona antiform (Austell–Frolona anticlinorium of *Crawford and Medlin* [1973]), where the antiform terminates and plunges beneath the surface at Austell, Georgia. Metasedimentary rocks within the antiform were intruded semiconcordantly by the Austell gneiss at or near the peak of metamorphism in the area. First generation (F_1) folds were coincident with this peak estimated at approximately 600°C and exhibit a well-developed axial planar schistosity (S_1). These folds are northeast trending, isoclinal, and overturned to the northwest.

Second generation (F_2) folds developed after peak metamorphism and produced overturned to upright northeast trending flow folds. F_2 folding is coaxial but not coplanar with F_1 and is responsible for the outcrop patterns observed at the northeast terminus of the Austell-Frolona antiform. There, S_1 foliations in the Austell gneiss bend around the nose of the F_2 fold [*Abrams and McConnell*, 1981*a*].

The third and fourth fold events were less intense than F_1 and F_2 and are not well developed throughout the entire study area. Third generation (F_3) folds also trend to the northeast but at a lesser angle than first and second genera-

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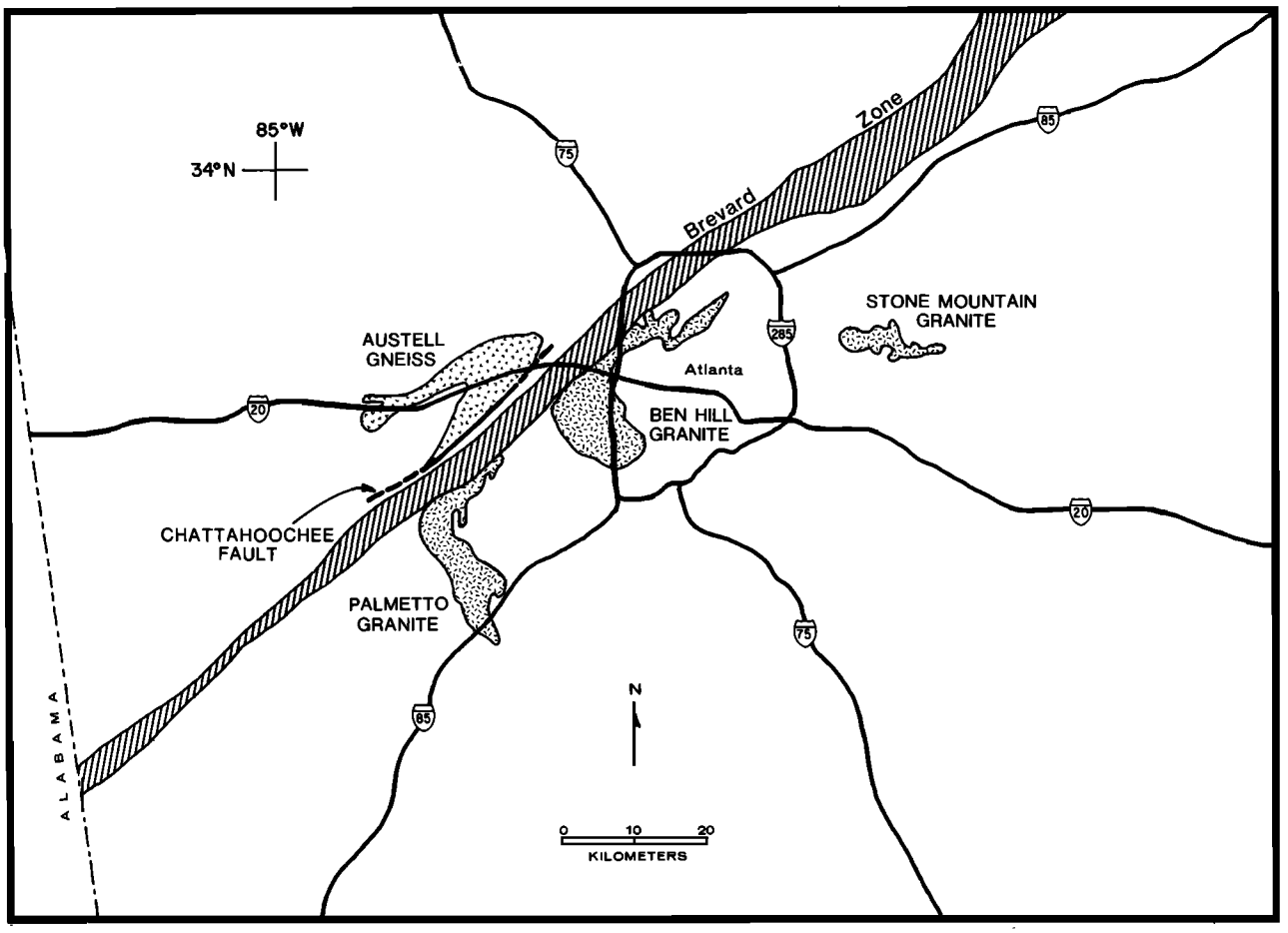


Fig. 1. Map of central-western Georgia centered on the Atlanta area and illustrating the location of gneissic and granitic units.

tion folds. This causes development of interference patterns in the Austell gneiss and surrounding rocks, as seen on the northwest limb of the gneiss. Fourth generation (F_4) folds are characterized by gentle, upright, and open warps which trend to the northwest.

Faulting. Rocks in the Austell-Frolona antiform, including the Austell gneiss, lie in fault contact with rocks of the Sandy Springs Group to the southeast. A reverse fault, the Chattahoochee fault [*Hurst,* 1973; *Crawford and Medlin,* 1973] has juxtaposed rocks of the Sandy Springs Group [*Higgins and McConnell,* 1978] with those in the Austell-Frolona antiform.

Age. The age of the Austell gneiss is only approximately known. *Dallmeyer* [1978] using $^{40}Ar/^{39}Ar$ methods has indicated that peak metamorphism in the Inner Piedmont, just southeast of the Brevard fault zone, occurred at approximately 365 m.y. *Abrams and McConnell* [1981*b*] have suggested that the Austell gneiss was intruded at or near peak metamorphism but was also affected by the D_1 (F_1) event. The unit was later further affected by the F_2-F_3 deformational events, but such events have not affected the undeformed Ben Hill and Palmetto granites to the southeast which are ~325 m.y. in age (ages based on U-Pb dates, *Atkins et al.,* [1981]) (Figure 1). This places limits on the age of the Austell gneiss between ~365 and ~325 m.y.

PREVIOUS WORK

Anisotropy of magnetic susceptibility (AMS). Magnetic susceptibility (K or χ, volume or mass susceptibility, respectively) is a symmetrical second-rank tensor coefficient relating an induced magnetic moment to the inducing magnetic field [*Nye,* 1969]. It is commonly quoted in terms of susceptibility per unit volume or mass in SI units and therefore is a dimensionless Systeme International (SI) number (equivalent to MKSA Practical Rationalized units), or as Gauss/oersted when using CGS Classical Unrationalized emu (see *Hanna* [1977] for a detailed discussion of MKS units used in magnetics, and *Nye* [1969] for further elaboration in terms of magnetic susceptibility measurements).

AMS data for any rock specimen are conventionally expressed as triaxial ellipsoids whose major, intermediate, and minor axes represent the corresponding maximum (K_a), intermediate (K_b), and minimum (K_c) susceptibility magnitudes and azimuths (magnitude ellipsoid of *Nye* [1969]). Bulk susceptibility can then be calculated from these magnitudes and is given as

$$\bar{K} = (K_a+K_b+K_c)/3 \quad (1)$$

For most rock specimens containing magnetite, it has been shown that the total susceptibility ellipsoid shape expresses the preferred alignment of included ferrimagnetic mineral grains [e.g., *Khan,* 1962; *Graham,* 1966], and therefore the AMS of such samples can be used as an indicator of petrofabrics in the rock [*Graham,* 1954].

The AMS for various types of metamorphic rocks, including metasediments, gneisses, and coals, has been reported by a number of different workers [e.g., *Balsley and Buddington,* 1960; *Khan,* 1962; *Graham,* 1966; *Hrouda and Janak,*

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1971; *Ellwood,* 1979*a*]. These authors are in agreement concerning the following: (1) K_c (minimum) susceptibility azimuths are aligned normal to the plane of foliation in the rock (at the pole to foliation), (2) K_c axes align parallel with the axis of maximum compression (uniaxial or biaxial) in the rock, (3) K_a (maximum) susceptibility azimuths parallel lineations in the rock and lie in the foliation plane, (4) K_a azimuths lie along the direction of fold axes, (5) K_a axes align normal to the direction of major compression, and (6) K_a axes are oriented parallel with the direction of maximum strain (extension). These effects generally result from plastic deformation, either before lithification of sediments or while the rock is held at high temperatures and pressures during the metamorphic process.

Remanent magnetization (RM). While the AMS for rock samples is contributed by the larger magnetic grains, the stable RM for the same samples resides in the very fine magnetic fraction [*Graham,* 1966; *Ellwood,* 1979*b*]. It has been shown that these two distributions of grain sizes can both yield useful but different information for the same samples, the larger grains yielding petrofabric information about the rock and site and the fine grains yielding a stable site RM consistency [e.g., *Ellwood and Whitney,* 1980; *Ellwood et al.,* 1980]. During metamorphic events, when rocks are brought to conditions of medium to high temperatures and pressures, the magnetic properties in such rocks change. Under certain conditions, magnetite mobility is high at relatively low temperatures (~400°C) due to high solubility of iron in chloride solutions [*Whitney et al.,* 1979]. The result may be the acquisition of a chemical RM (CRM) at relatively low temperatures (curves for the reacquisition of an RM due to heating for relatively long periods of time are given by *Pullaiah et al.* [1975]).

METHODS

Field. A total of 63 cores were drilled with a gasoline-powered portable drill from two quarries, two fresh flat-lying outcrops, and six road cuts in the Austell gneiss (Figure 2) and oriented using a Brunton compass. Cores were selected which avoided any unusual lithologies, highly weathered areas, or flame cut surfaces and were separated laterally by an average distance of greater than 3 m.

Laboratory. The cores of ~2.5-cm diameter, were sliced into lengths of ~2.2 cm to conform to an optimum length to diameter ratio range of 0.82–0.90 [*Banerjee and Stacey,* 1967; *Noltimier,* 1971]. The AMS was measured for six samples from each site using an automated low-field torque magnetometer (the method has been described by *King and Rees* [1962]).

The natural remanent magnetization (NRM) was measured for all samples using a Schonstedt spinner magnetometer. A stepwise af demagnetization was performed at 10, 20, 40, 60, and 80 mT. Selected samples were thermally demag-

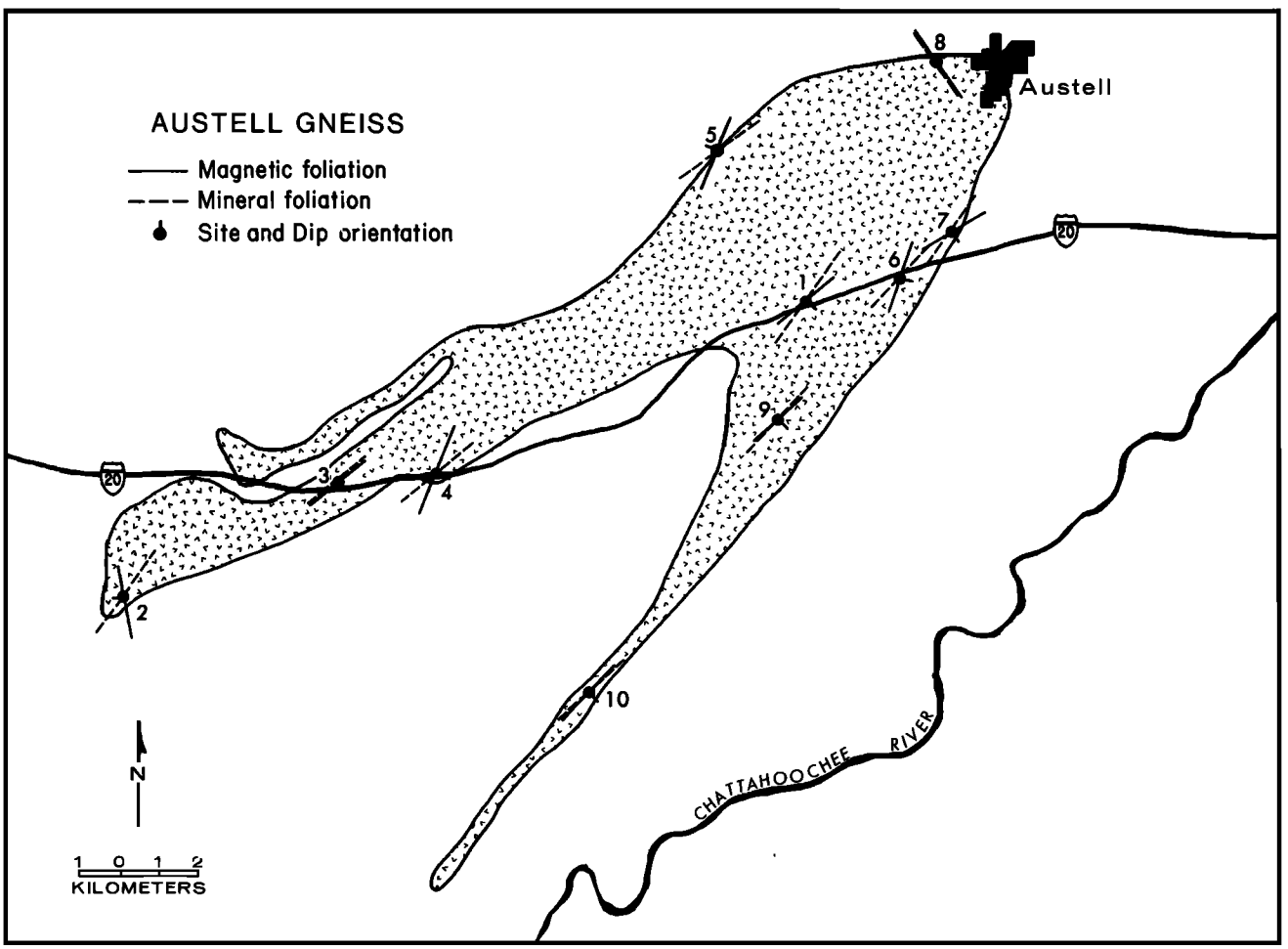


Fig. 2. Outcrop pattern of the Austell gneiss, with site locations numbered and labeled. Solid bars are the strike of the magnetic foliation planes. Dashed bars are the strike of the mineral foliation planes. Dip direction is illustrated by short bars.

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netized at 100°C steps to 700°C, and the RM was measured after each run. One sample from each site was given an isothermal remanent magnetization (IRM) in fields of 20, 40, 60, 80, 100, 200, 300, and 400 mT, and the magnetization measured. Polished sections of 10 samples from 6 sites were examined under magnification up to 1000× using an Olympic petrographic microscope with oil immersion lens.

Statistical. Since the AMS of a sample is a symmetric physical property with the induced moment always in the same sense as the applied field, a principal AMS axis defines two directions on a unit sphere. Further, a random scatter of the axial directions about some preferred azimuth defines a bimodal distribution of induced moment orientations. This is very different from the unimodal scatter of RM vectors. Therefore, principal axis mean directions were calculated using the eigenvalue method (summarized by *Mark* [1973]), which is independent of any one direction assigned to a principal axis.

Within-site scatter about an AMS axial mean is represented by

$$S_a \text{ or } S_c = \lambda_1/(\lambda_1 + \lambda_2 + \lambda_3) \quad (2)$$

and λ_1, λ_2, and λ_3 are the eigenvalues of the mean AMS azimuth. S_a or S_c can range from 1.0, representing a cluster of identical axial orientations and a unidirectional ellipsoid (highly prolate), to 0.333 for a random scatter of orientations whose representation quadric is a sphere.

RESULTS

Typical af and thermal demagnetization data are illustrated in Figures 3–5. Based on these and the results for samples from all sites, it was decided that the 40-mT af demagnetization data best characterize the stable RM recorded in the Austell. Data from each site were evaluated for internal consistency, and anomalous directions were not considered in calculating the site mean directions given in Table 1. Eliminated samples represent either anomalous polarities or an attempt to improve the precision of the data. As a rule, if three or more samples from a site yielded similar directions (±10°), then the mean of these directions was used as a standard for the site. If the great circle distance between this standard and any one sample was >50°, the sample direction was considered anomalous and the direction was not used in compiling Table 1. Two RM means are reported for the unit (Table 1), one utilizing eight sites, excluding sites 6 and 8 with poor within site precision, $\kappa<6.0$ (κ is from *Fisher* [1953]), and the other excluding two more sites, AS 9 and AS 10 which exhibit anomalous directions.

Mean AMS data for each site are given in Table 2. Also reported are the precision indicators, S_a for K_a data and S_c

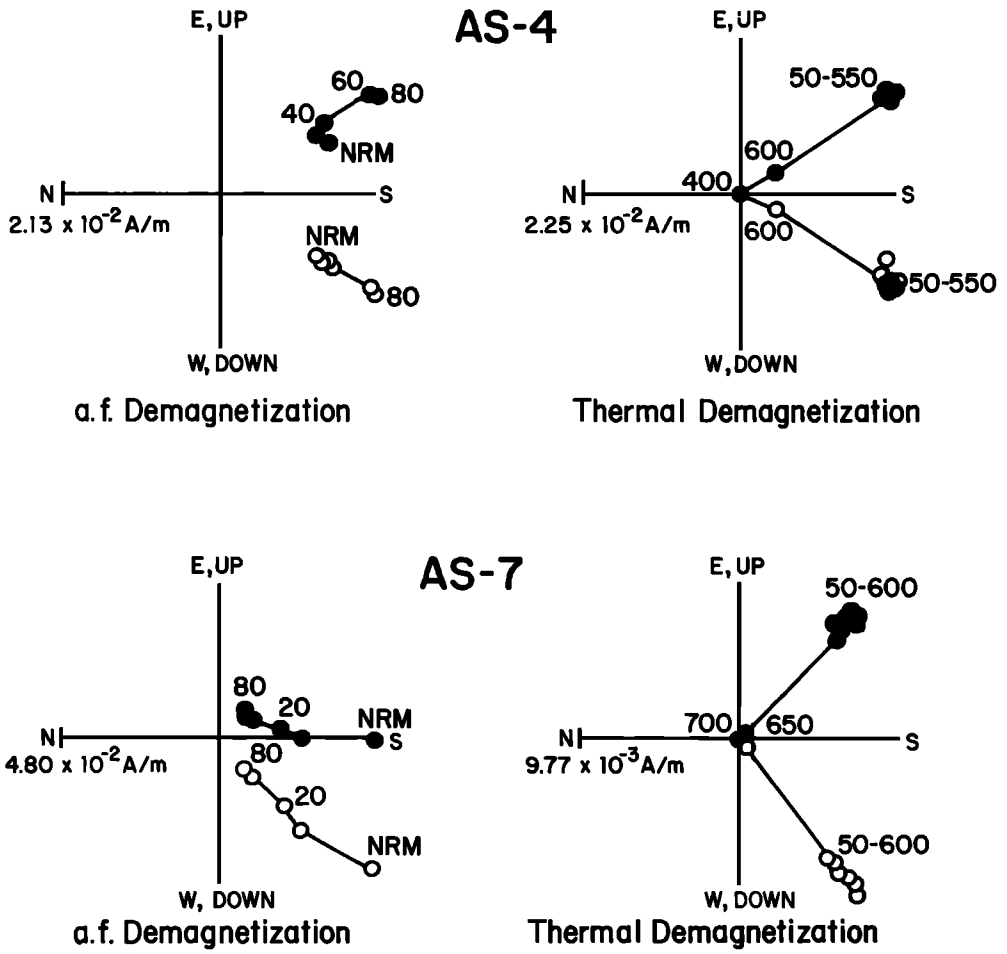


Fig. 3. *Zijderveld* [1967] vector plot of samples AS 4 and AS 7 after af demagnetization to 80 mT and thermal demagnetization to 700°C. Solid circles represent projections on a horizontal plane. Open circles are plotted on a vertical plane. Intensity of magnetization in A/m.

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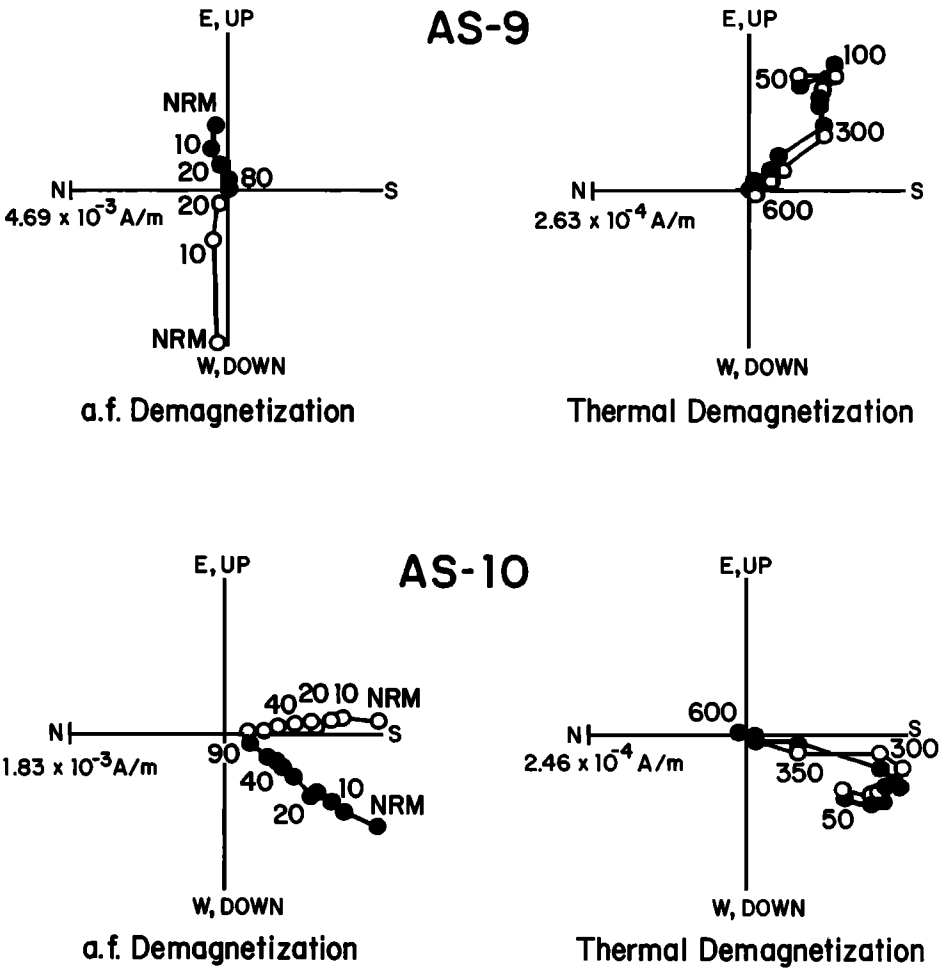


Fig. 4. *Zijderveld* [1967] vector plot of samples AS 9 and AS 10 after af demagnetization to 80 mT and thermal demagnetization to 600°C. Symbols as in Figure 3. Above 600°C samples became cracked and swollen. Directions appeared spurious and were not used.

for K_c data. A typical data distribution for a single site is given in Figure 6.

Discussion

Magnetic mineralogy. Examination of polished sections for samples from the Austell show magnetite, ilmeno-hematite, and some very slight occurrences of pyrrhotite grains. Magnetite grains range in size from 10 μ down to the resolution of the microscope, less than 2 μ. Some samples are weathered and discolored to shades of red as a result of the production of very fine (<1 μ) hematite grains.

Rock magnetism. IRM experiments for samples from each site were conducted to characterize the magnetite grain size distribution and contribution of hematite to the magnetization of those samples. Typical results are given in Figure 7. From these data it is inferred that relatively fine-grained magnetite is present in most samples, which show near saturation occurring at fields of 0.1 T. Also evident in all samples is a steady increase of IRM intensity with increasing field, which is indicative of hematite. For example, AS 2 has actually been discolored red as a result of oxidation of magnetite to fine-grained hematite. The slope of each IRM curve from 0.3 to 0.7 T (Figure 7) is indicative of the magnitude of this increase. The af and thermal demagnetization curves after IRM acquisition are given in Figure 8. The more hematite-rich sample AS 2 exhibits a slow decrease in intensity with increasing af field, while AS 5 exhibits typical fine-grained magnetite behavior with initial removal of an apparent viscous RM by fields of 20 mT and then very little decrease in intensity with continued demagnetization. Nor-

TABLE 1. Austell Remanent Magnetic Site Mean Results

	N_1/N_2	D	I	J	κ
Site					
AS 1	6/4	165.1	13.8	8.72×10^{-4}	13.8
AS 2	6/3	162.3	24.3	5.43×10^{-3}	34.8
AS 3	6/5	169.6	32.9	4.96×10^{-3}	11.9
AS 4	6/6	147.2	30.0	1.48×10^{-1}	36.9
AS 5	6/4	149.0	41.2	3.05×10^{-2}	9.4
AS 6*	6/6	100.4	67.4	1.39×10^{-3}	2.4
AS 7	8/6	131.5	36.4	1.42×10^{-2}	180.4
AS 8*	7/6	136.7	6.1	5.34×10^{-3}	1.8
AS 9†	6/4	206.7	21.8	1.42×10^{-3}	38.1
AS 10†	6/4	227.1	13.3	5.26×10^{-2}	6.0
AS Mean 1	8	170.5	30.1	—	7.5
AS Mean 2	6	154.8	30.4	—	27.4

Site location is from Figure 2. N_1 is the number of samples drilled at each site; N_2 is the number of samples within each site used in calculating the site mean. D, I, and J and the RM declination and inclination in degrees, and intensity of magnetization after demagnetization, in amperes per meter, respectively; $\kappa = (N - 1)/N - R)$ [*Fisher*, 1953].

*Sites rejected in calculating AS Mean 1 and 2.

†Sites rejected in calculating AS Mean 2 (see text).

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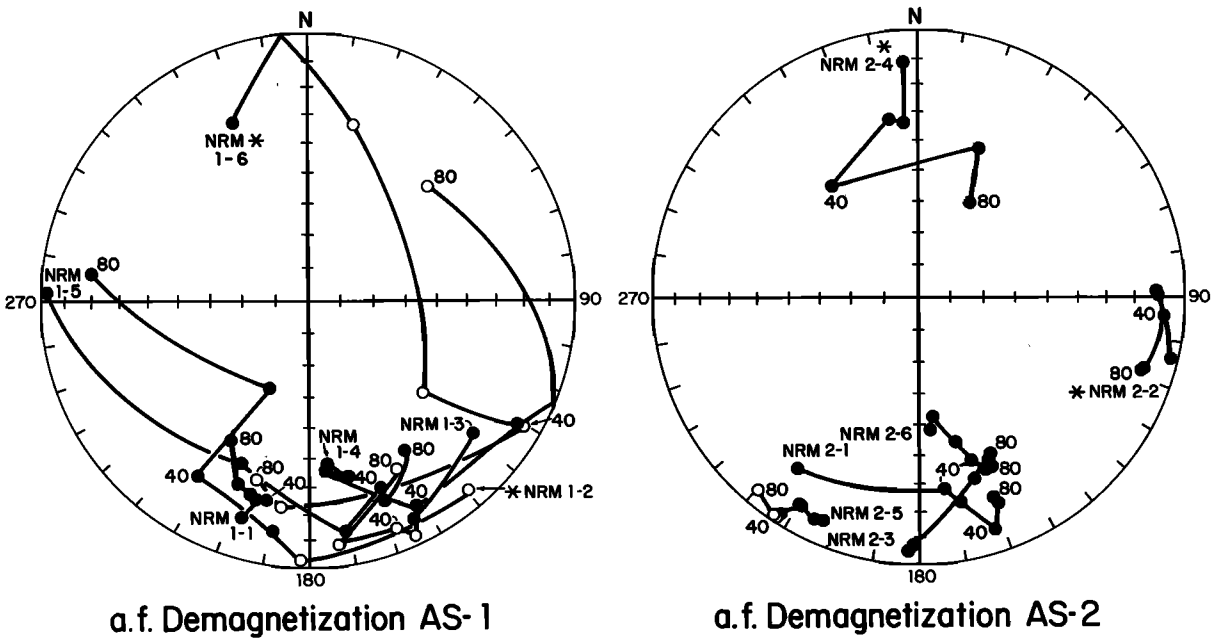


Fig. 5. Equal area plots of RM directions after af demagnetization for samples from two Austell sites. Demagnetization fields in mT. The asterisks indicate samples exhibiting spurious directions.

malized intensity of magnetization after thermal demagnetization decreases uniformly to ~700°C and is consistent for all samples. These results indicate that fine-grained magnetite is probably the main carrier of the stable RM exhibited by most Austell sites.

Typical *Zijderveld* [1967] vector plots of af and thermal demagnetization data show a wide range of behavior (Figures 3–4). Decay toward the origin with af demagnetization is observed for many samples (Figure 4), but in some samples, very little change or a trend away from the origin occurs with af or thermal demagnetization until temperatures >500°C are reached (Figure 3). Clearly, stability in at least some of these samples is controlled, in part, by the presence of hematitie or very stable, fine-grained, single-domain magnetite. Some RM vectors, upon progressive af demagnetization, move directly away from the origin after an initial but slight change due to the removal of a small viscous component of magnetization (e.g., AS 4, Figure 3). This change away from the origin is interpreted to indicate the removal of a reversed (relative to the RM of the sample) overprint, carried in lower coercivity (<70 mT) magnetic components.

The susceptibility, which is often equated with magnetite content [e.g., *Nettleton,* 1971], is relatively constant throughout the Austell (Table 2). Since susceptibility is moderately high ($\bar{K} = 2.9 \times 10^{-4}$ in SI units), these data as well as the optical, IRM, and thermal demagnetization data indicate the presence of relatively fine-grained magnetite in all sites as well as significant amounts of hematite in most sites.

Austell fabric. It has been shown that the magnetic fabric of unaltered dikes in Iceland [*Ellwood,* 1979*c*] and in dikes cutting granites in the southern Piedmont (B. B. Ellwood, unpublished data, 1978) is parallel to the dike orientation. Altered dikes of greenschist metamorphic grade (~500–525°C, J. A. Whitney, personal communication,

TABLE 2. Austell Anisotropy of Magnetic Susceptibility Site Mean Data

		K_a			K_c				
	N	D	I	S_a	D	I	S_c	V	$\bar{K} \times 10^{-4}$
Site									
AS 1	6	45.5	7.1	0.930	129.2	−39.5	0.967	45.1	1.51
AS 2*	6	26.8	−21.7	0.595	57.8	74.9	0.720	53.4	1.88
AS 3	6	54.2	15.7	0.934	141.5	−10.4	0.926	36.9	1.27
AS 4	4	75.9	71.7	0.837	110.8	−14.4	0.983	60.4	4.67
AS 5*	6	100.9	−46.4	0.870	113.3	45.7	0.963	58.0	1.59
AS 6	6	40.5	5.9	0.832	119.3	−57.9	0.870	50.5	3.60
AS 7	6	19.6	−15.4	0.824	120.5	−37.6	0.955	49.7	8.40
AS 8*	7	327.1	0.4	0.683	55.1	−40.2	0.673	42.3	1.70
AS 9	6	48.4	7.3	0.971	136.9	−12.3	0.980	43.2	1.72
AS 10	6	44.5	0.1	0.836	133.5	−46.2	0.980	48.4	3.06
Means	···	43.9	12.2	···	127.9	−31.6	···	48.9	2.94

K_a is the maximum; K_c is the minimum AMS; $\bar{K}$ is the bulk susceptibility; see text for an explanation of the precision of cluster parameter $S_{a,c}$ (1.0 − perfect cluster; 0.333 = random cluster); V is computed from the relationship $\sin^2 V = (K_b - K_c)/(K_a - K_c)$ and is an indicator of deformation [*Graham,* 1966]; site, N, D, I, as in Table 1.

*Sites rejected in calculating means.

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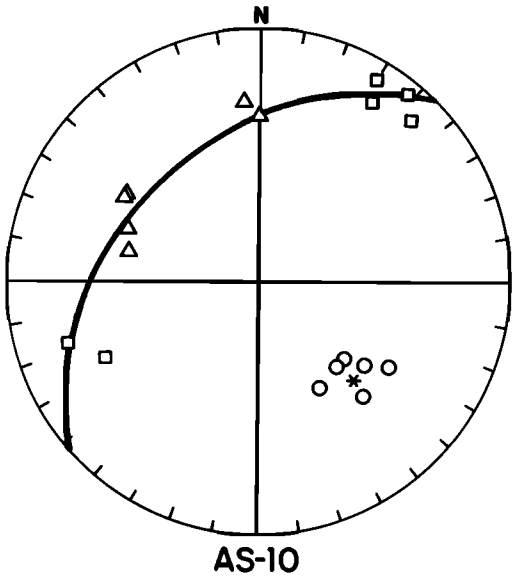


Fig. 6. Equal-area plot of the AMS results for site 10 in the Austell gneiss. Solid line represents the orientation of the magnetic foliation plane. Open squares, K_a; open triangles, K_b; open circles, K_c; asterisk, mean K_c direction.

1981), however, have their fabrics reoriented to conform with the metamorphosed host rock fabric [*Whitney et al.*, 1978; B. B. Ellwood, unpublished data, 1979]. The magnetic and mineral fabrics, measured for 10 sites in the Austell gneiss, are illustrated in Figure 2. Clearly, both fabrics are similar and both generally exhibit parallel, near margin conformity. These data are interpreted to indicate magnetite recrystallization during deformation.

The K_a and K_b AMS principal axes for a single site are used here to define the within-site maximum susceptibility plane. For sites which have tightly clustered K_c axes, this maximum susceptibility plane represents the orientation of the magnetic foliation for the site (e.g., Figure 6). The magnetic foliation plane is easiest to quantify by first determining the mean K_c azimuth which lies at the pole to the plane. The strike of this plane is a horizontal direction within the plane and normal to the K_c mean. The dip of the plane is equal to 90 minus the absolute value of the mean K_c inclination. S_c (Table 2) is an indicator of the precision of the K_c axial cluster. It is apparent that while a visual close similarity exists for most sites between magnetic foliation and mineral foliation, the magnetic foliation for sites 2, and 8 is poorly defined ($S_c < 0.8$, Table 2). Sites 2, 5, and 8 exhibit anomalous magnetic fabric orientation relative to the overall

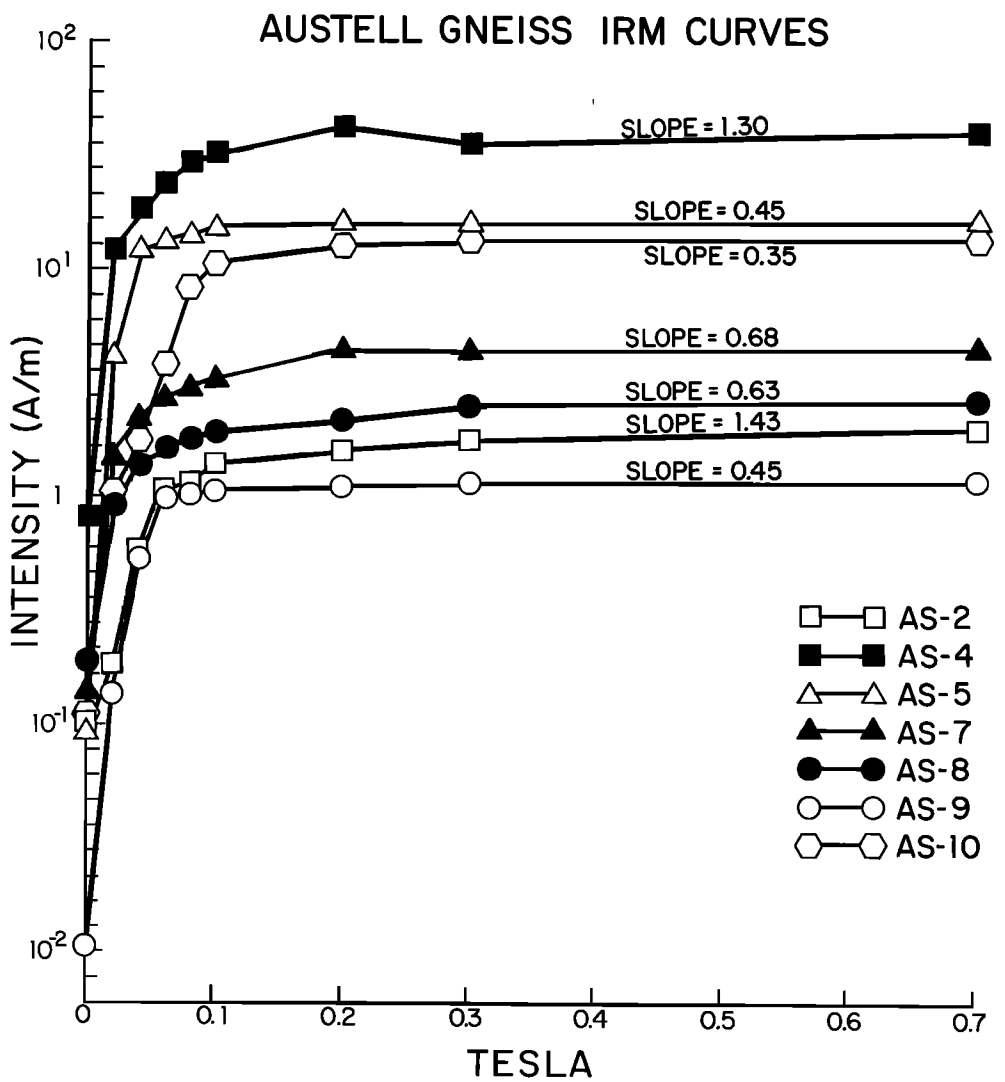


Fig. 7. Isothermal remanent magnetization (IRM) curves for seven Austell samples. Numbers represent the site.

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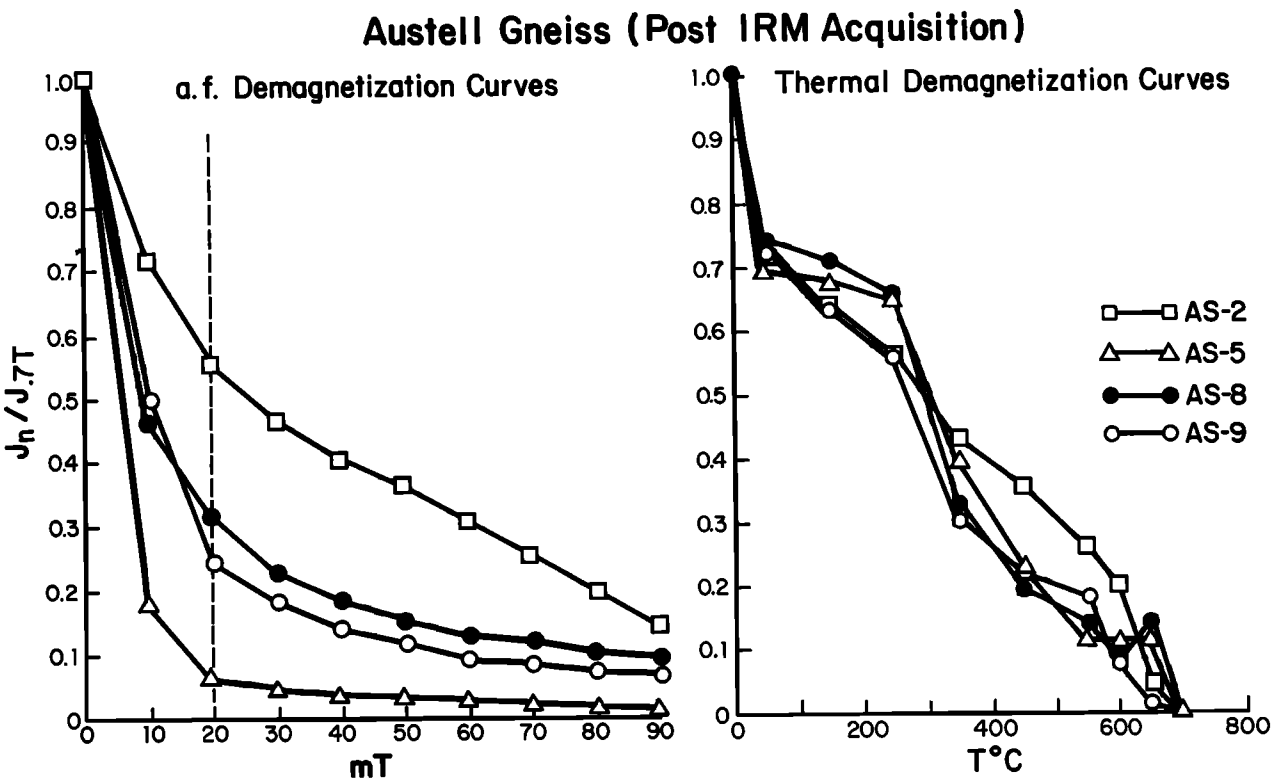


Fig. 8. Normalized af and thermal demagnetization plots after samples have been given an IRM in a 0.7-T field. Data for four of the seven samples reported in Figure 7.

trends at all other sites. The poor precision for sites 2 and 8 may be the result of site 2 being partially affected by F_3 and F_4 deformational events and site 8 being located in the nose of the fold. Anomalous directions at site 5 may result from the sites close position (<20 m) to the contact between the Austell gneiss and the bounding country rock. For example, this author has observed that fabric very near the contact of the Elberton granite with the country rock is controlled by local contact irregularities. This control decreases rapidly with distance from the margin and appears to no longer be a significant factor at distances from the contact of >20 m or so. In some cases, such as for site 5, the magnetic foliation may be better defined than is the mineral foliation due to a poorly developed nearly indistinguishable mineral fabric.

The magnetic fabric is generally oblate in nature, as indicated by *Graham's* [1966] V (Table 2). This is true for seven of the 10 Austell sites. Where $V>45$ and S_c for K_c axial clusters is high ($S_c>0.9$), deformation is not intense, but the magnetite foliation is well defined. Three sites exhibit a pronounced prolate fabric ($V<45$) with two of these yielding well grouped K_a axial directions ($S_a>0.9$) indicative of a well-defined magnetic lineation at these sites. In general, however, lineation is not as well defined as is the magnetic foliation.

Compressive stresses. Since it has been established that K_c (minimum) principal axes will develop parallel to the direction of maximum compressive stress [e.g., *Graham*, 1966], the data in Table 2 can give an estimate of compressive stress orientation during the period of magnetite grain mobility and realignment in the Austell gneiss. While stresses may vary locally within the unit depending on many factors, we infer that the K_c mean calculated for all sites, excluding anomalous directions (K_c mean in Table 2), yields a NW-SE oriented regional compressive stress direction present during the development of the S_1-S_2 foliations at, and shortly following, peak metamorphism at ≤365 m.y. Folding developed in the presence of this stress field. The orientation of the fold axis can be represented by the mean K_a AMS azimuth (magnetic lineation), which exhibits a NE-SW direction (K_a mean, Table 2). Plunge, represented by the downward magnetic lineation trend, is to the NE at homologous ~12°. These observations are similar to the outcrop patterns exhibited by the unit (Figures 1 and 2) and the mineral lineations in the gneiss which are plunging at ~18 to 27° to the NE.

The plane of magnetic foliation (plane of maximum strain) within the unit is normal to the inferred NW-SE directed compressive stress direction but inclined at an angle of ~50° to the SE (K_c mean, Table 2). This may be due to (1) a gentle warping of the area during the F_3-F_4 deformational events, (2) an overriding thrust sheet producing an inclined compressive stress, or (3) late stage tectonic tilting of the unit resulting from motion along the Brevard Fault Zone immediately to the SE of the Austell.

Tilt corrections. Ellwood and Whitney [1980] have developed a case for tilting of the Elberton granite in NE Georgia, based on the outcrop pattern of the body, xenolith concentrations, and the extreme uniformity of the magnetic fabric. Some of the same arguments are now applicable to the Austell. The magnetic fabric of the unit closely resembles the outcrop pattern of the body (Figure 2) and appears to have been developed during folding and deformation rather than as a result of tectonic rotation. The unit does, however, resemble a plunging anticlinal body in outcrop pattern, and the inclination of ~12° to the NE suggested for the Austell magnetic lineations may be the result of regional doming in

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TABLE 3. Fold Test Data

		RM		Magnetic Foliation		RM		Mineral Foliation	
	N	D_1	I_1	Strike	Dip	D_2	I_2	Strike	Dip
AS 1	4	162.6	−35.7	N50.8E	50.5SE	181.9	32.7	N35E	48NW*
AS 2	3	167.3	19.5	N32.2W	15.1NW*	152.4	−30.7	N35E	80SE
AS 3	5	161.2	−49.1	N51.5E	79.6SE	294.9	59.6	N50E	74NW*
AS 4	6	142.7	−19.5	N20.8E	75.6SE	125.9	−40.6	N50E	72SE
AS 5	4	210.6	66.6	N23.3E	44.3NW*	316.2	77.7	N53E	57NW*
AS 7	6	136.0	−15.0	N59.5E	52.4SE	121.9	−10.1	N35E	50SE
AS 9*	4	223.5	−13.5	N46.9E	77.7SE	219.2	−7.7	N43E	74SE
AS 10*	4	228.2	10.5	N43.5E	43.8SE	231.2	6.7	N47E	55SE
Mean	8	176.9	−7.6	$\kappa = 2.8$		183.0	15.9	$\kappa = 1.6$	
Mean Less	6	157.5	−8.7	$\kappa = 3.3$		140.4	−34.7	$\kappa = 3.7$	

$D_{1/2}$, $I_{1/2}$, are tilt-corrected D and I based on the assumption of an originally horizontal 1 magnetic/2 mineral foliation plane; D, I, N, site, κ, as in Table 1.
* Sites rejected from the second mean.

the area as discussed above. While it is difficult to evaluate the validity of using the AMS data for the purposes of making tilt corrections, especially in the absence of well-defined sedimentary bedding planes, such a tilt correction is the only one indicated for the body by either the AMS or geologic field data.

Fold test. The folded Austell gneiss provides the opportunity to perform a fold test on the RM site mean data. The magnetic and mineral foliation data have been used to correct individual RM site mean directions (Table 3) by assuming that the measured foliation in the body was initially horizontal and has been rotated away from the horizontal plane during the F_1 and/or F_2 folding of the gneiss. It is important to note, however, that such an assumption, while reasonable, may not be correct. Mean directions calculated from the corrected directions are given for all sites in Table 3. Precision for these mean directions as indicated by κ is poor for both the magnetic and mineral fabric corrected data. This indicates that either the magnetic directions measured for the Austell gneiss were acquired after the unit was folded (our preferred explanation), or that the correction factors used in tilting individual sites back to the horizontal were incorrect and therefore the assumption of original horizontality of the mineral foliation for this unit is incorrect.

RM directions. Site mean directions for the 10 Austell sites generally exhibit fair precision ($\kappa > 5.0$) and in the case of site AS 7, excellent precision ($\kappa = 180.4$). These mean directions, less anomalous or poor precision sites AS 6, 8, 9, and 10, are plotted in Figure 9. Also plotted in Figure 9 are the normal and reversed mean directions for the Middle Devonian-Upper Permian (mid to late Paleozoic) North American paleomagnetic pole position ($\phi = 120.6$, $\theta = 47.1$, Table 4) calculated from summary data reported by *Van der Voo et al.* [1980]. The field direction during this time varied only slightly. A great circle (Figure 10) has been drawn to pass through the Devonian-Permian mean directions, and the mean direction calculated for the Austell data ($\phi = 126.2$, $\theta = 34.4$; Table 4). The Austell direction is slightly steeper than the mid to late Paleozoic North American direction. There are at least four possible explanations for this discrepancy.

1. During magnetization of the Austell, the magnetic field direction at the site was slightly different from the reported mid to late Paleozoic direction.

2. The area may have been tectonically moved, postmagnetization, and the Austell magnetic direction may have deviated from an original primary direction as a result. One possible tilt correction has been performed on the data, based on the assumption of a plunging antiform, discussed in an earlier section. Corrected results (Table 4) do not improve the correspondence between the North American Poles and the pole calculated from the mean Austell direction.

3. The direction in the Austell is spurious.

4. Cooling and magnetic acquisition in the gneiss may have occurred over long periods of time, during which reversals occurred in the Earth's magnetic field direction. Both polarities may have been recorded in individual samples due to varying blocking temperatures. If the field direction changed slightly during this period, then a deflection along a great circle toward the polarity which the earth's

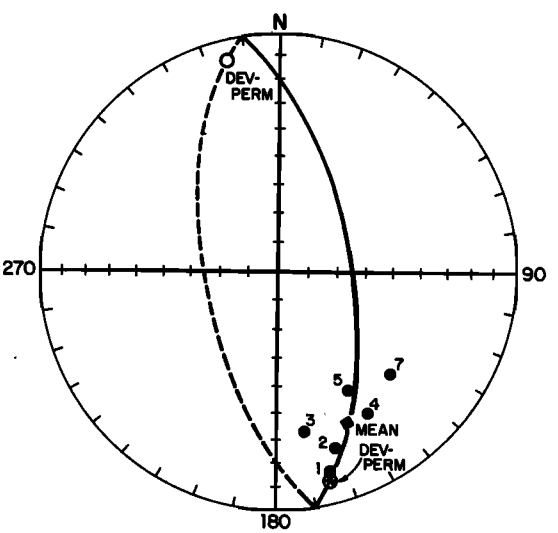


Fig. 9. Equal-area projection of the Austell gneiss RM individual site (solid circle) and site mean (solid diamond) directions. Open circle, Devonian-Permian normal and circled star, Devonian-Permian reversed directions [*Van der Voo et al.*, 1980]. Solid and dashed lines represent the plane containing the Devonian-Permian directions and the Austell mean direction. Open symbols and the dashed great circles are lower hemisphere plots; solid symbols and the solid great circle are plotted on the upper hemisphere.

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TABLE 4. Austell Gneiss Unit Mean Remanent Magnetic Results

Site	N	$\bar{D}$	$\bar{I}$	ϕ	θ	dp	dm
AS Mean 1	8	170.5	30.1	128.3	39.2	43.0	43.2
AS Mean 2	6	154.8	30.4	126.2	34.4	6.6	13.2
M DEV-U PERM		~165	~11	120.6	47.1	...	...
Tilt-Corrected as Mean 2	6	143.3	34.2	132.2	28.9	8.7	15.0

Magnetic Lineation Azimuth is N43.9E; dip = 12.2SE. N, number of sites used in calculating the mean direction; $\bar{D}$, mean declination; $\bar{I}$, mean inclination; ϕ, θ are paleomagnetic pole, longitude and latitude; dp is the error in determining the great circle distance of the pole from the sample site; dm is an error in declination which corresponds to a displacement of the pole, normal to the pole–sample site great circle.

field maintained for the shortest length of time, would be observed. Such a deflection is represented schematically in Figure 10, where changes in the field direction are represented by angular limits determined for the Middle Devonian–Upper Permian mean directions from *Van der Voo et al.* [1980] and percentages represent the relative time the earth's field spent in each polarity. These results in conjunction with demagnetization data suggest that the geomagnetic field was represented by both polarities (reversed dominating) during the magnetization of the Austell gneiss.

Evaluating RM data from gneissic or granitic units is usually difficult since such units generally yield directions with poor RM precision. Further, tilt estimates can only be approximate when distinct geological criteria to indicate tilting, such as sedimentary bedding planes, can not be correlated to the unit. When evaluating the pole position for the Austell, errors due to (1) the large oval of confidence defined for the pole position, (2) unrecognized tilt events, and (3) the streaked great circle distribution of reversed RM directions toward the mid to late Paleozoic normal direction (Figure 9) observed for the unit must be considered. Because the Austell pole is poorly constrained, the data can accommodate several possible interpretations. First, the unit may have acquired an RM direction during long-term cooling while the Austell was located, relative to stable North America, essentially in its present location. Second, the possibility of a thrust of unknown magnitude translating the unit to the northwest cannot be excluded from an interpretation of the data. Third, a large-scale translation of >1000 km along a transcurrent fault to the northeast, similar to that recognized for the Maritime Provinces of North America by *Kent and Opdyke* [1978], is consistent with the Austell data.

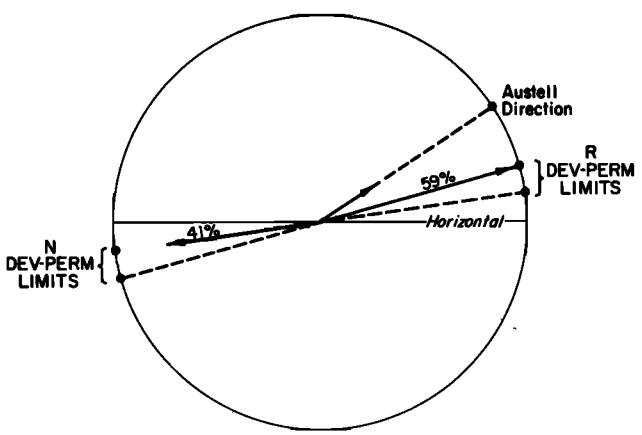


Fig. 10. Cross section of Figure 9 along the Devonian-Permian-Austell mean great circle. See text for a discussion.

Point three is based on the fact that a paleolatitude calculated for the Austell gneiss (Table 4) indicates acquisition of magnetization at 10°–11° of latitude south of the expected mid-to-late Paleozoic location for the Austell. Such a translation is difficult to justify if the geology of the area is considered, since the Austell lies to the northwest of the major Piedmont tectonic boundaries along which such movement might have occurred. One can overcome these geologic objections, however, with the argument that the crustal block originally containing the Austell now lies to the southeast of the Brevard fault zone. After moving northeastward to its present position, the surface rocks of the block, which included the Austell gneiss, were thrust to the northwest, bringing the Austell to its present location. In favor of point three above, is the similarity between the paleopole position calculated for the Austell gneiss, and other mid to late Paleozoic poles from the Maritime Provinces of North America [e.g., *Brecher et al.*, 1974].

CONCLUSIONS

1. The magnetic fabric exhibited by the Austell gneiss closely approximates the mineral fabric, suggesting that both were acquired during the F_1 or F_2 deformational events which have affected the body.
2. A paleomagnetic fold test performed on the unit was unsuccessful. This result is consistent with a postfolding RM acquisition sometime between 325 and 365 m.y.
3. Our results indicate that the RM was acquired during magnetite recrystallization or cooling from metamorphic temperatures (~600°C).
4. RM acquisition in the gneiss appears to have occurred over a relatively long period of time, with both reversed and normal polarities represented in the magnetic signature of the unit.
5. At least three explanations for the observed Austell paleopole position exist. From simple to complex these are (1) magnetic acquisition occurred essentially at the present location of the unit, (2) thrusting to the northwest after RM acquisition brought the Austell to its present location, or (3) the Austell acquired its magnetization, was transported to the northeast >1000 km by a left-lateral transcurrent fault system, and was later thrust to the northwest to the present location of the body.

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